

ASSIGNMENT – 10

ADVANCED PROGRAMMING

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ROLLNO- CSB24045

```
1 class Address:
2     def __init__(self, street, city, zipCode):
3         self.street = street
4         self.city = city
5         self.zipCode = zipCode
6
7     def display(self):
8         return f"{self.street}, {self.city} - {self.zipCode}"
9
10
11 class Student:
12     def __init__(self, name, age, address):
13         self.name = name
14         self._age = None
15         self.age = age
16         self.address = address
17         self.courses = []
18
19     @property
20     def age(self):
21         return self._age
22
23     @age.setter
24     def age(self, value):
25         if value <= 0 or value > 120:
26             raise ValueError("Invalid age")
27         self._age = value
28
29     def add_course(self, course):
30         self.courses.append(course)
31
32     def display(self):
33         print(f"Name: {self.name}")
34         print(f"Age: {self.age}")
35         print(f"Address: {self.address.display()}")
36         print(f"Courses: {self.courses}")
37
```

```

39 class ScholarshipStudent(Student):
40     def __init__(self, name, age, address, scholarshipAmount):
41         super().__init__(name, age, address)
42         self.scholarshipAmount = scholarshipAmount
43
44     def display(self):
45         super().display()
46         print(f"Scholarship Amount: {self.scholarshipAmount}")
47
48
49 addr = Address("Main Street", "Tezpur", "784001")
50
51 s1 = Student("Rahul", 20, addr)
52 s1.add_course("Math")
53 s1.add_course("Physics")
54
55 s2 = ScholarshipStudent("Anita", 22, addr, 50000)
56 s2.add_course("Chemistry")
57
58 s1.display()
59 print()
60 s2.display()

```

OUTPUT-

```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 2
c:\Users\VITU\OneDrive\Desktop\java>python -u "c:\Users\VITU\OneDrive\Desktop\java\student_system.py"
Name: Rahul
Age: 20
Address: Main Street, Tezpur - 784001
Courses: ['Math', 'Physics']

Name: Anita
Age: 22
Address: Main Street, Tezpur - 784001
Courses: ['Chemistry']
Scholarship Amount: 50000

c:\Users\VITU\OneDrive\Desktop\java>

```